Physics 603
Spring 2025
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Homework 6

Due Monday, April 7

1. (10 pts.) Problem 8.C from the textbook. Please replace "find the Dynkin coefficients ..." with "using the $q-p$ diagram (boxes), find all the positive roots." There should be 9 of those. Please use the following set of simple roots:

$$\begin{aligned} \alpha^1 &= (1,-1,0)\,, \\ \alpha^2 &= (0,1,-1)\,, \\ \alpha^3 &= (0,0,1)\,. \end{aligned}$$

2. (10 pts.) Consider the rank-3 Lie algebra of SO(6), using the following set of simple roots:

$$\begin{aligned} \alpha^1 &= (1,-1,0)\,, \\ \alpha^2 &= (0,1,-1)\,, \\ \alpha^3 &= (0,1,1)\,. \end{aligned}$$

The Cartan matrix is

$$A_{ji} = \frac{2\alpha^j \cdot \alpha^i}{(\alpha^i)^2} = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 2 \end{pmatrix}.$$

Consider the **second** fundamental representation of this algebra, corresponding to Dynkin coefficients $l^1 = 0$, $l^2 = 1$, $l^3 = 0$.
(a) Using the $q-p$ diagram, construct this representation and find its dimensionality. (All weights of this irrep are non-degenerate.)
(b) The highest-weight state of this representation (as a vector in the Cartan space) is $\mu^2 = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$. Find all the weights (as vectors in the Cartan space). As a check of your calculations, verify that all the weight vectors have the same length.